

Neighbour-Driven Gaussian Process Variational Autoencoders for Scalable Structured Latent Modelling

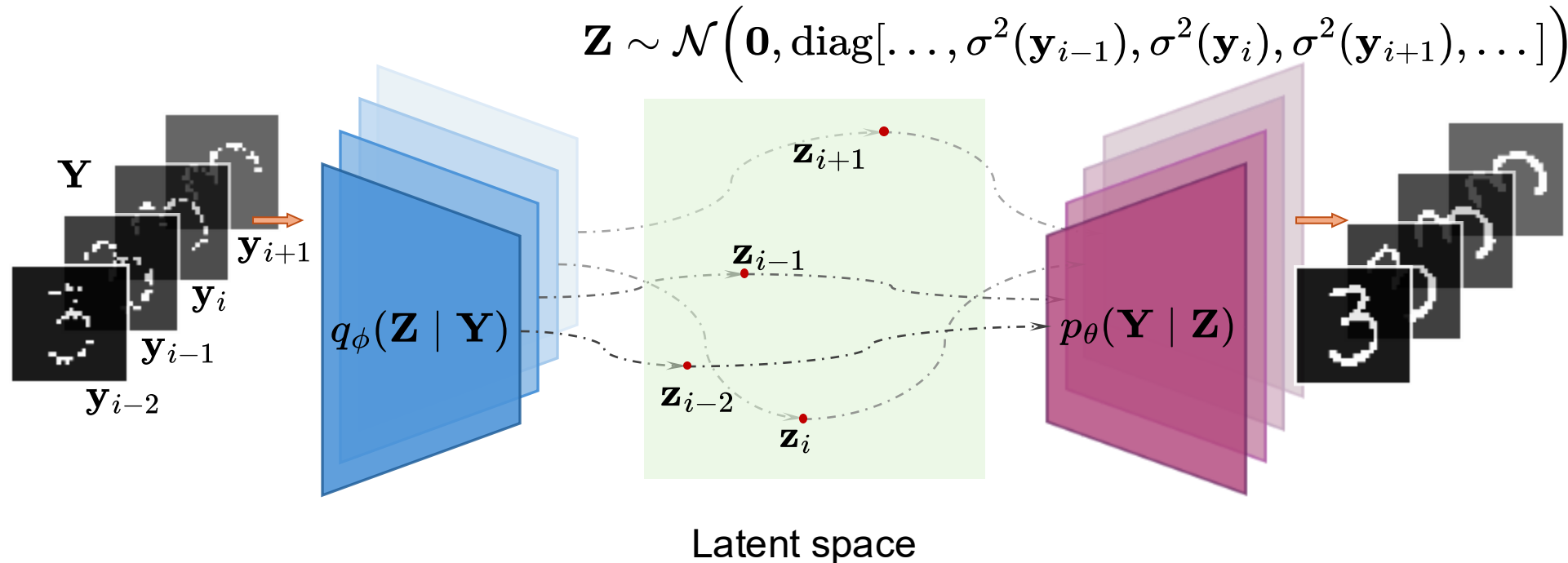
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VAE: Independent Latent Variables

- A standard VAE assumes a fully factorised Gaussian prior over the latent space for amortised variational inference
- This assumption ignores correlations of structured data Y with “timestamps” X

$$\mathbf{Y} = [\mathbf{y}_n]_{n=1}^N \in \mathbb{R}^{N \times K}, \quad \mathbf{X} = [\mathbf{x}_n]_{n=1}^N \in \mathbb{R}^{N \times D}$$



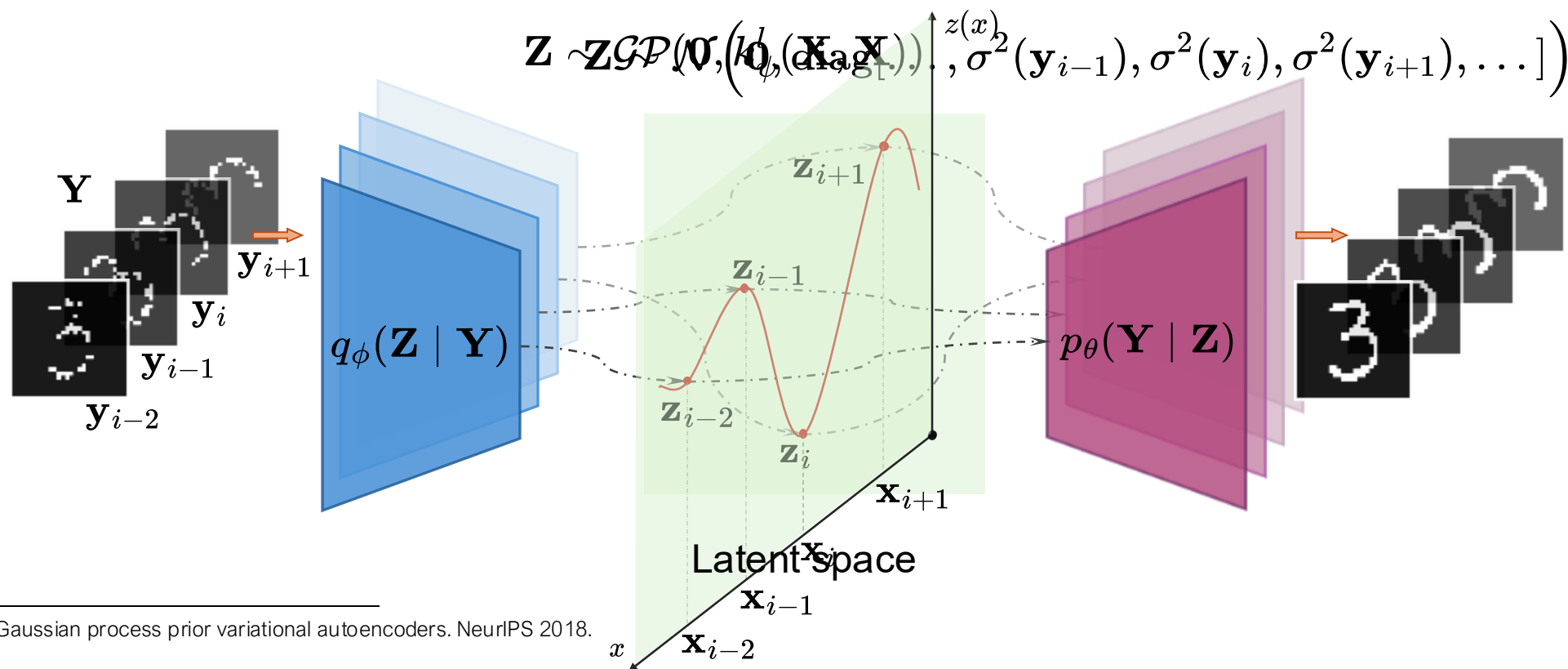
GPVAE: GP Prior in VAE but with Poor Scalability

- Build correlations via Gaussian process (GP) prior on data pairs $(\mathbf{X}, \mathbf{Y})$

$$\mathbf{Y} = [\mathbf{y}_n]_{n=1}^N \in \mathbb{R}^{N \times K}, \mathbf{X} = [\mathbf{x}_n]_{n=1}^N \in \mathbb{R}^{N \times D}$$

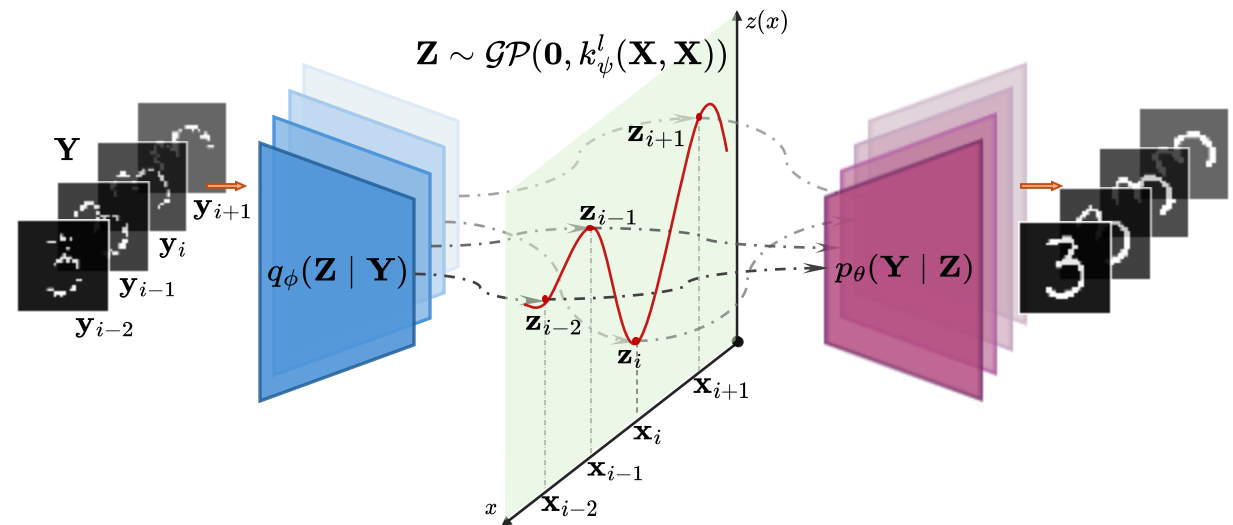
- ELBO scales poorly with $\mathcal{O}(N^3)$

$$\mathcal{L} = \mathbb{E}_{q_\phi(\mathbf{Z}|\mathbf{Y})} [\log p_\theta(\mathbf{Y} | \mathbf{Z})] - \text{KL} [q_\phi(\mathbf{Z} | \mathbf{Y}) || p_\psi(\mathbf{Z} | \mathbf{X})]$$



Previous Methods

- Specific kernel structures, such as certain Matérn / low-rank kernels
 - narrow kernel choices
- Using inducing points to approximate GP posterior
 - numerous inducing points
 - optimisation difficulties
- Sampling-based methods, such as SGHMC
 - longer runtime



Casale, F. P., et al. Gaussian process prior variational autoencoders. NeurIPS 2018.

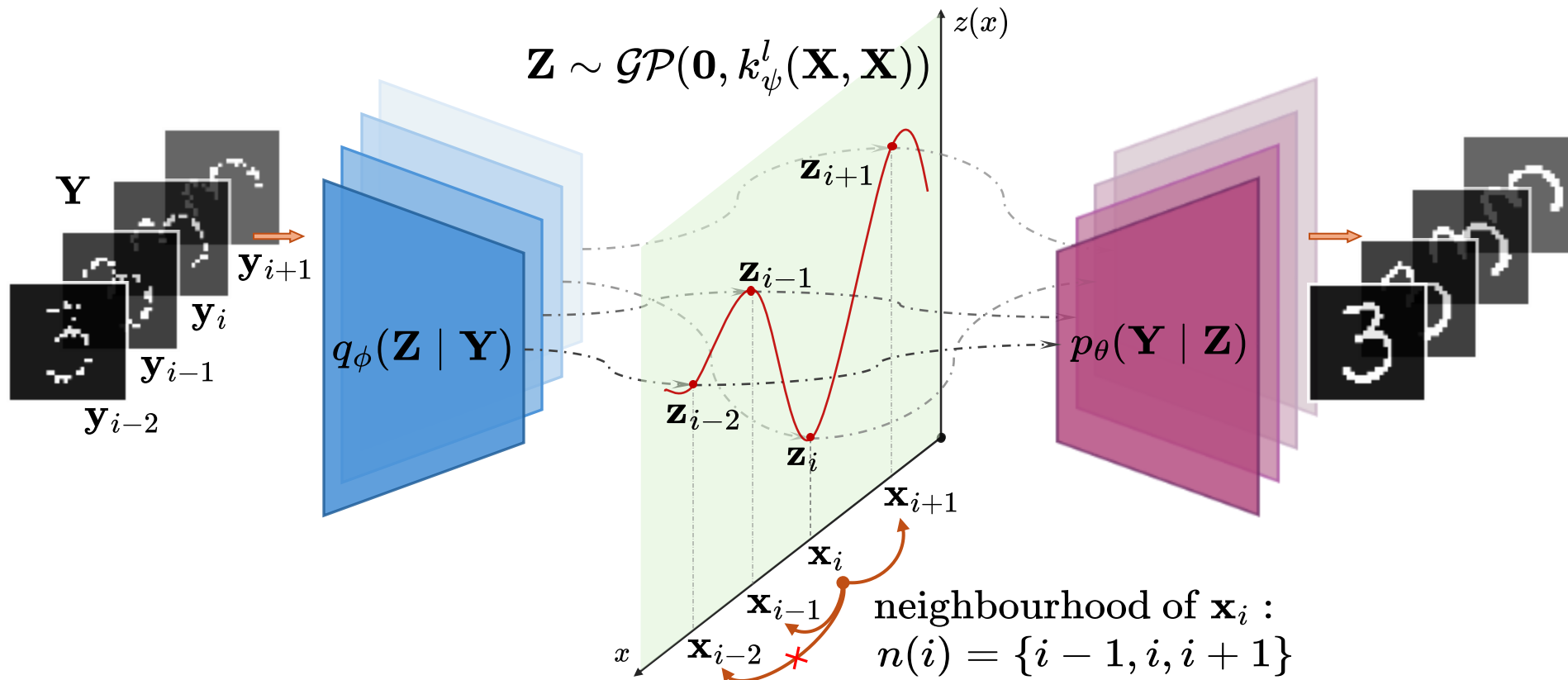
Zhu, H., et al. Markovian Gaussian process variational autoencoders. ICML 2023.

Jazbec, M., et al. Scalable Gaussian process variational autoencoders. AISTATS 2021.

Tran, B.-H., et al. Fully Bayesian autoencoders with latent sparse Gaussian processes. ICML 2023.

Key Idea: Talk Only to Nearest Neighbours

- Inspired by the “first law of geography” in recent Nearest Neighbour GP (NNGP)
- Focusing on a small set of nearest neighbours captures most of the essential correlation structures



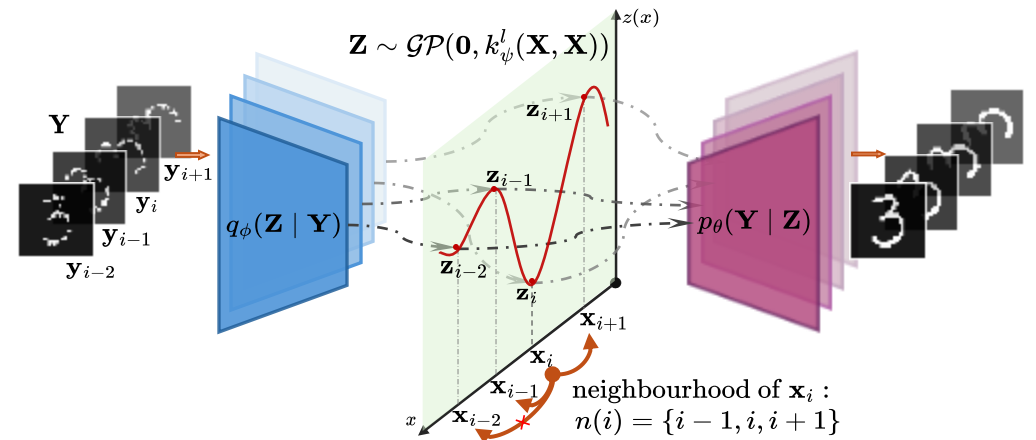
Two Neighbour-Driven Prior Approximations

- Hierarchical Prior Approximation (HPA): sparsify covariance via hierarchical masks

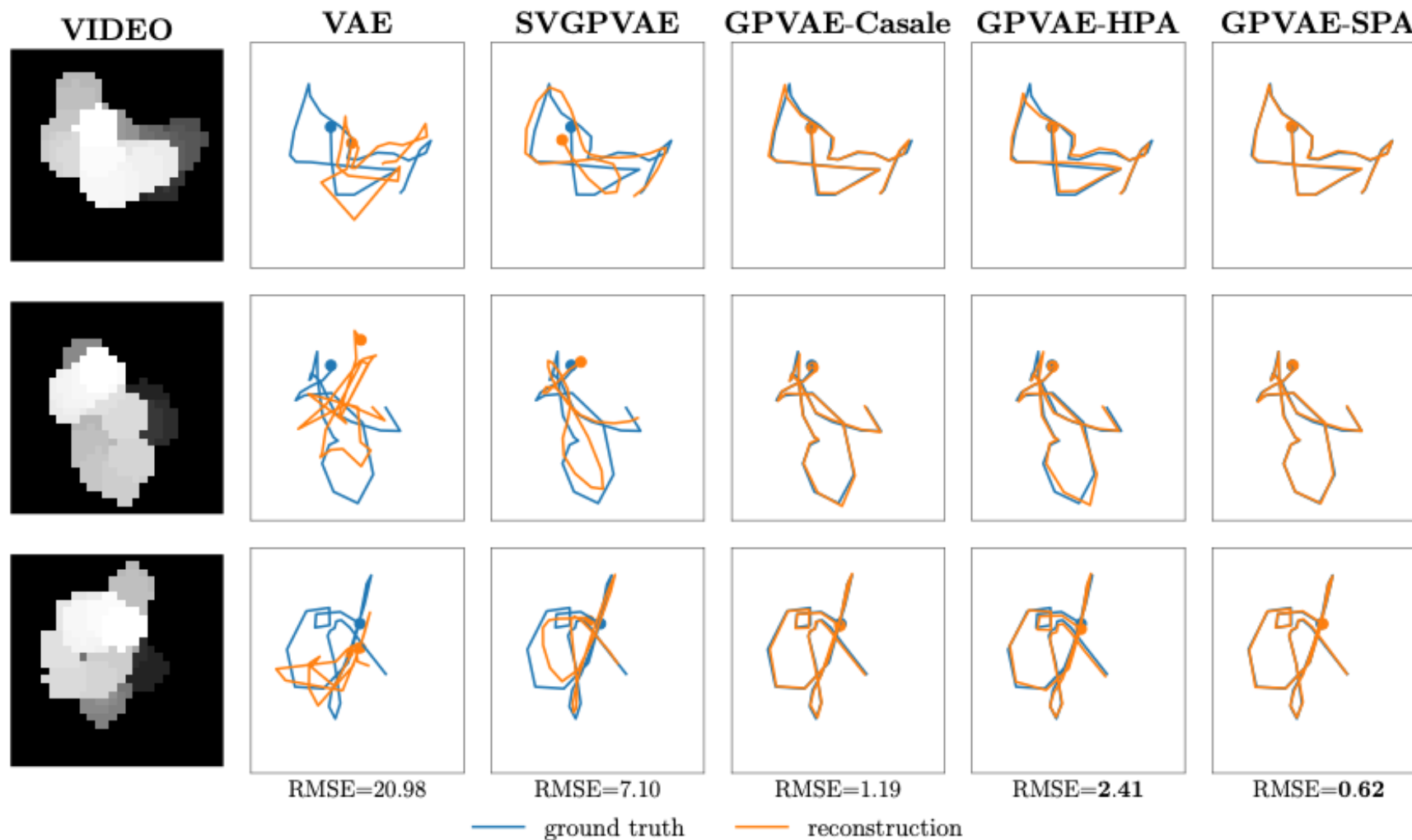
$$\mathcal{L}_{\text{HPA}} \approx \frac{N}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \left\{ \mathbb{E}_{q(\mathbf{z}_i | \mathbf{y}_i)} [\log p(\mathbf{y}_i | \mathbf{z}_i)] - \frac{1}{N} \text{KL} [q(\mathbf{Z}_{n(i)}) || p(\mathbf{Z}_{n(i)})] \right\}$$

- Sparse Precision Approximation (SPA): sparsify precision matrix via chained conditionals

$$\mathcal{L}_{\text{SPA}} \approx \frac{N}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbb{E}_{q(\mathbf{z}_i | \mathbf{y}_i)} [\log p(\mathbf{y}_i | \mathbf{z}_i)] - \frac{N}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \mathbb{E}_{q(\mathbf{z}_{n(j)})} \text{KL} [q(\mathbf{z}_j) || p(\mathbf{z}_j | \mathbf{Z}_{n(j)})]$$



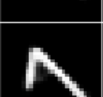
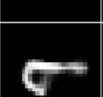
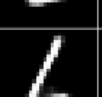
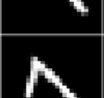
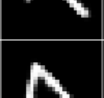
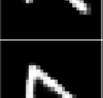
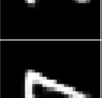
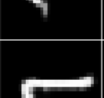
Experiments: Latent Representation in Moving Ball



Experiments: Impute Corrupted Frames in Rotated MNIST

Input										
VAE										
HI-VAE										
GPVAE-Diag (full batch)										
GPVAE-Band (full batch)										
MGPVAE										
GPVAE-HPA (ours)										
GPVAE-SPA (ours)										

Experiments: Generate Missing Frames in Rotated MNIST

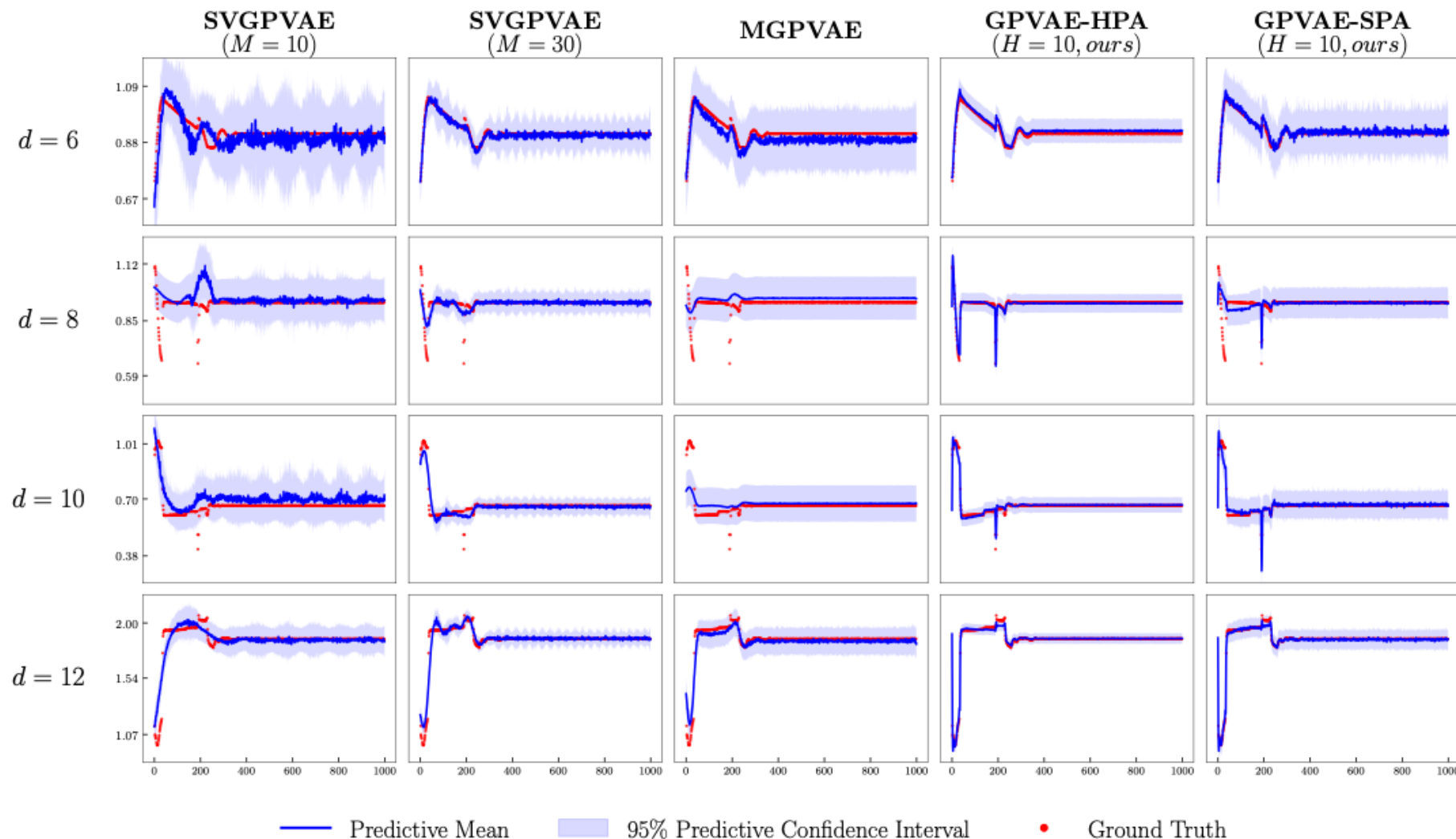
Input																			
SVGPVAE ($M = 10$)																			
SVGPVAE ($M = 30$)																			
LVAE ($M = 10$)																			
LVAE ($M = 30$)																			
MGPVAE																			
GPVAE-HPA ($H = 10, ours$)																			
GPVAE-SPA ($H = 10, ours$)																			

Experimental Results: Rotated MNIST

Table 1: Performance of different models for tasks of the corrupted and missing frames imputation on Rotated MNIST.

	Models	NLL	RMSE	Training Time (s/epoch)
Corrupted Frames	VAE (Kingma & Welling, 2014)	0.193 ± 0.038	0.333 ± 0.011	75.741 ± 0.660
	HI-VAE (Nazabal et al., 2020)	0.114 ± 0.002	0.222 ± 0.002	76.969 ± 0.594
	GPVAE-Diag (Fortuin et al., 2020)	0.094 ± 0.000	0.199 ± 0.000	127.614 ± 2.758
	GPVAE-Banded (Fortuin et al., 2020)	0.146 ± 0.001	0.189 ± 0.001	768.347 ± 24.297
	MGPVAE (Zhu et al., 2023)	0.090 ± 0.001	0.197 ± 0.001	110.772 ± 0.651
	GPVAE-HPA-H5 (ours)	0.095 ± 0.000	0.199 ± 0.000	137.860 ± 0.326
	GPVAE-SPA-H5 (ours)	0.096 ± 0.000	0.202 ± 0.000	121.185 ± 0.817
Missing Frames	SVGPVAE-M10 (Jazbec et al., 2021)	87.711 ± 0.484	6.122 ± 0.015	8.280 ± 0.111
	SVGPVAE-M30 (Jazbec et al., 2021)	77.792 ± 0.592	5.572 ± 0.030	9.180 ± 0.188
	LVAE-M10 (Ramchandran et al., 2021)	73.887 ± 0.748	5.578 ± 0.022	28.919 ± 0.253
	LVAE-M30 (Ramchandran et al., 2021)	73.545 ± 1.035	5.558 ± 0.024	29.063 ± 0.122
	MGPVAE (Zhu et al., 2023)	72.217 ± 0.192	5.496 ± 0.005	25.103 ± 0.103
	GPVAE-HPA-H10 (ours)	71.264 ± 1.263	5.521 ± 0.046	9.819 ± 0.064
	GPVAE-SPA-H10 (ours)	69.538 ± 0.664	5.412 ± 0.026	9.358 ± 0.083

Experiments: Generate Missing Timestamps in MuJoCo



Experiments: Generate Missing Timestamps in MuJoCo

Table 2: Results of conditional generation at missing timestamps on MuJoCo dataset.

Models	NLL	RMSE	Training Time (s/epoch)
SVGPVAE-M10 (Jazbec et al., 2021)	-1.512 ± 0.032	0.0583 ± 0.003	44.000 ± 5.477
SVGPVAE-M30 (Jazbec et al., 2021)	-1.071 ± 0.175	0.0391 ± 0.001	47.893 ± 3.170
MGPVAE (Zhu et al., 2023)	-1.708 ± 0.057	0.0398 ± 0.002	57.291 ± 1.141
GPVAE-HPA-H10 (ours)	-2.335 ± 0.032	0.0222 ± 0.001	37.000 ± 3.793
GPVAE-SPA-H10 (ours)	-1.715 ± 0.159	0.0378 ± 0.007	39.403 ± 2.755

Experiments: Impute Missing Locations in SPE10

Table 3: Imputation results of various models on SPE10 dataset.

Models	NLL	RMSE	Training Time (s/epoch)
MOGP-M1000 (Hensman et al., 2013)	6.034 ± 0.003	1.230 ± 0.000	3.820 ± 0.021
VNNGP-H20 (Wu et al., 2022)	1.052 ± 0.000	0.683 ± 0.000	2.490 ± 0.084
VAE (Kingma & Welling, 2014)	3.436 ± 0.015	1.298 ± 0.002	1.416 ± 0.020
HI-VAE (Nazabal et al., 2020)	0.752 ± 0.034	0.647 ± 0.008	1.448 ± 0.011
SVGPVAE-M100 (Jazbec et al., 2021)	2.025 ± 0.020	0.979 ± 0.006	4.755 ± 0.097
SGPBAE-M20 (Tran et al., 2023)	1.335 ± 0.106	0.898 ± 0.125	1585.6 ± 21.3
GPVAE-HPA-H20	0.735 ± 0.007	0.638 ± 0.007	4.019 ± 0.036
GPVAE-SPA-H20	0.736 ± 0.004	0.638 ± 0.004	3.365 ± 0.117

Conclusion

- HPA & SPA: covariance vs precision sparsity in GPVAE
- No specific kernel structures
- Do not rely on inducing points
- Amortised variational inference with faster training

Code: <https://github.com/shixinxing/NNGPVAE-official>